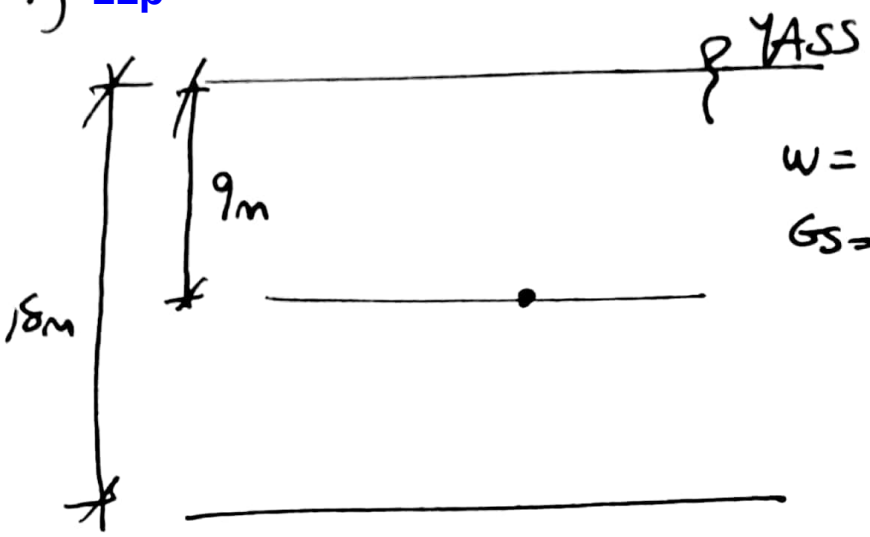
i) 11p



$G_s \cdot w = S \cdot e$

$2.67 \cdot 0{,}25 = 1 \cdot e$

$e = 0.6675$ 3p

$$\gamma_d = \frac{(G_s + e)\cdot \gamma_w}{1+e} = \frac{(2.67 + 0{,}66{,}75)\cdot 9{,}81}{1{,}6675} = 19{,}63\ kN/m^3$$ 2p

9m derinlikte →

$\sigma = 19{,}63 \cdot 9 = 176.71\ kN/m^2$ 2p

$u = 9 \cdot 9{,}81 = 88{,}29\ kN/m^2$ 2p

$\sigma' = 88.42\ kN/m^2$ 2p

ii) 9p

200 — %25 / %75 → ince daneli zemin

w_L = %58 → CH yada MH

P_L = %38

IP = 20 %

↓

MH → USCS 5p

A-7-5 → AASHTO 4p

8p a) $q = k \cdot \Delta H \cdot N_f / N_d = 4{,}8 \cdot 10^{-6} \cdot 400 \cdot 5/11 = 8{,}72 \cdot 10^{-4}\ cm^3/s$ 8p

12p b) $H_A = 4 - 3 \cdot (4/11) = 2{,}91\ m$ 2p

$H_B = 4 - 8 \cdot (4/11) = 1.091\ m$ 2p

$H_C = 4 - 9 \cdot (4/11) = 0.727\ m$ 2p

$H_A = \frac{U_A}{\gamma_w} + z_A \Rightarrow 2{,}91 = \frac{U_A}{9{,}81} - 3 \Rightarrow U_A \cong 58\ kPa$ 2p

2p $U_B = 59{,}7\ kPa$

2p $U_C = 26.75\ kPa$

③ 15p

a) $\sigma_{v_0}' = 6.18{,}54 + 3.5.(20-9.81) \cong 146.6\ kPa$ 4p

$\sigma_{v_0}' + \Delta\sigma = 3.18{,}54 + 3{,}5.(20-9.81) + \frac{200.20.20}{(20+6.57).(20+6{,}5)}$ 4p

$\cong 205\ kPa$ 3p

$$f_c = \frac{c_c}{1+e_0}.H.\log\left(\frac{\sigma_{v_0}'+\Delta\sigma}{\sigma_{v_0}'}\right) = \frac{0.125}{1+0.78}.7.\log\left(\frac{205}{146.6}\right)$$

$= 0.0715m$ 4p

5p

b) $T = \frac{c_v.t}{(H_d)^2} \rightarrow \frac{0{,}35.10}{(3.5)^2} = 0.287 \rightarrow U = \%60$ 3p

NC kil rain $\rightarrow 0{,}60.\ 0{,}0715 = 0{,}043m$ 2p

3) Suya doygun örselenmemiş aşırı konsolide kil numuneleri üzerinde yapılan üç eksenli basınç deneyi sonuçları aşağıda verilmiştir.

Deney	Hücre Basıncı (kPa)	Deviatorik Gerilme (kPa)
1.CD Deneyi	120	190
2.CD Deneyi	240	360

Buna göre;

a) Kil zeminin drenajlı kayma mukavemeti parametrelerini hesaplayınız.

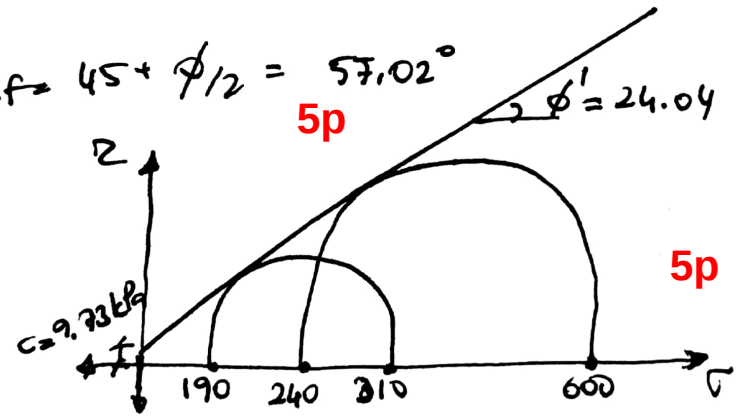
b) CD deneyinde göçme anında göçme düzleminin yatayla yaptığı açıyı (α_k) hesaplayınız.

c) CD deneyinden deneyi sonuçlarına göre göçme anındaki Mohr Dairesini ve göçme zarfını çiziniz.

10p a) $\phi' = 24.49$ 5p $\quad c' = 6.43$ kPa 5p

5p b) $\alpha_f = 45 + \phi/2 = 57.02^\circ$ 5p

5p c)



5p

Arazi	Dolgu	Lab
$e=0.82$	$V=10000m^3$	$\gamma k_{maks.}=18.5 kN/m^3$
$G_s=2.65$	$V_{SU}=?$	$w_{opt}=\%13$
$V_{arazi}=?$		$G_s=2.65$

$RC=\%95$

a) **10p**

$$RC=\frac{\gamma k_{dolgu}}{\gamma k_{lab}}=\frac{\gamma k_{dolgu}}{\gamma k_{lab}}=\frac{\gamma k_{dolgu}}{\gamma_{maks}}=0.95$$

$$\gamma k_{dolgu}=17.575 kN/m^3$$

2p

$$W_{karazi}=W_{kdolgu}$$

$$\gamma k_{arazi}.V_{arazi}=\gamma k_{dolgu}.V_{dolgu}$$

$$\gamma k_{arazi}=\frac{G_s.\gamma_w}{1+e}=\frac{2.65.9,81}{1+0.82}=14.283 kN/m^3$$

3p

$$14.283.V_{arazi}=17,575.10000$$

$$V_{arazi}=12304.8\ m^3$$ **5p**

b) **10p**

$w_{opt}=\%13$

$W_{dolgu}=175750 kN$

$$\frac{M_{SU}}{M_S}=\frac{W_{SU}}{W_k}=0.13 \qquad W_{SU}=0,13.175750$$

$W_{SU}=22847.5 kN$ **5p**

$V_{SU}=$ $22847.5/9.81=2329\ m^3$ **5p**